

The Solar System

The Solar System consists of the Sun and everything that orbits it, including eight planets, dozens of moons, asteroids, comets, and dwarf planets. The Sun contains about 99.8 percent of the total mass of the Solar System and its gravity holds everything in orbit around it.

The four inner planets, Mercury, Venus, Earth, and Mars, are called terrestrial planets because they have solid, rocky surfaces. Mercury is the smallest planet and closest to the Sun, with extreme temperature swings between day and night. Venus is the hottest planet due to a thick atmosphere of carbon dioxide that traps heat through a runaway greenhouse effect.

Earth is the only known planet to support life, with liquid water covering about seventy percent of its surface. Mars, often called the Red Planet due to iron oxide on its surface, has the largest volcano in the Solar System, Olympus Mons, and evidence suggests it once had flowing water billions of years ago.

Beyond Mars lies the asteroid belt, a region containing millions of rocky bodies left over from the formation of the Solar System. The four outer planets, Jupiter, Saturn, Uranus, and Neptune, are called gas giants or ice giants because they lack solid surfaces and are composed mostly of hydrogen, helium, and other gases or ices.

Jupiter is the largest planet in the Solar System, with a mass greater than all other planets combined. It has a famous storm called the Great Red Spot that has raged for centuries. Saturn is known for its spectacular ring system, made up of ice and rock particles ranging in size from tiny grains to large boulders.

Uranus is unique because it rotates on its side, with an axial tilt of about ninety-eight degrees, likely caused by a massive collision early in its history. Neptune, the farthest known planet from the Sun, has the strongest winds in the Solar System, reaching speeds of over two thousand kilometers per hour.

Beyond Neptune lies the Kuiper Belt, home to dwarf planets like Pluto, which was reclassified from a full planet to a dwarf planet in 2006 by the International Astronomical Union. The Solar System also contains countless comets, icy bodies that develop glowing tails of gas and dust when they approach the Sun.